Business Requirements Document

Meta Ads Performance Analysis

Executive Summary

This document outlines the business requirements for developing a comprehensive performance tracking and analysis report for advertising campaigns running on Meta platforms (Facebook and Instagram). The solution will provide actionable insights to optimize campaign performance, improve ROI, and enable data-driven budget allocation decisions.

Business Objective

The primary business need is to establish a centralized performance tracking and analysis dashboard for Meta advertising campaigns that enables the marketing team to:

Primary Objectives

- **Identify the most effective platform** by comparing Facebook vs. Instagram campaign performance
- **Track campaign ROI** and optimize budget allocation across platforms, ad types, and audiences
- **Understand audience engagement patterns** across demographics (age, gender), geography, and time
- **Optimize ad creative performance** by analyzing ad type effectiveness
- **Detect temporal trends** to inform campaign scheduling and budget pacing

Expected Business Outcomes

- Improved campaign ROI through data-driven budget reallocation
- Enhanced targeting precision based on demographic and geographic insights
- Optimized ad scheduling based on hourly and seasonal performance trends
- Reduced cost per acquisition (CPA) through continuous performance monitoring
- Faster decision-making with real-time, interactive dashboards

Scope - In Scope

Platforms:

- Facebook (paid advertising campaigns only)
- Instagram (paid advertising campaigns only)

Out of Scope- Excluded Platforms:

- Messenger ads
- Audience Network
- WhatsApp Business
- Third-party advertising platforms

Key Performance Indicators (KPIs)

Conversion & Performance Metrics

KPI	Definition	Formula	Primary Use Case
Purchases	Number of purchases made after seeing ads	Count of event_type = Purchase	Measure conversion volume
Conversion Rate	Percentage of clicks that resulted in purchases	(Purchases ÷ Clicks) × 100	Funnel efficiency analysis
Purchase Rate	Percentage of impressions that resulted in purchases	(Purchases ÷ Impressions) × 100	Conversion from reach measurement

Engagement Metrics

KPI	Definition	Formula	Primary Use Case
Impressions	Number of times ads were displayed	Count of event_type = Impression	Measure reach and visibility
Clicks	Number of times users clicked ads	Count of event_type = Click	Measure engagement intent
Shares	Number of times ads were shared	Count of event_type = Share	Viral engagement measurement
Comments	Number of user comments on ads	Count of event_type = Comment	User sentiment & feedback
Engagements	Total interactions (Clicks + Shares + Comments)	Clicks + Shares + Comments	Overall engagement volume
CTR (Click-Through Rate)	Percentage of impressions that resulted in clicks	(Clicks ÷ Impressions) × 100	Ad effectiveness measurement
Engagement Rate	Percentage of impressions that resulted in engagements	(Engagements ÷ Impressions) × 100	Overall ad appeal

Budget Metrics

KPI	Definition	Formula	Primary Use Case
Total Budget	Total spend allocated to campaigns	Sum of campaigns.total_budget	Cost analysis
Avg. Budget per Campaign	Average budget allocation per campaign	Total Budget ÷ Campaign Count	Budget distribution analysis

Charts Requirements:

1. Target Gender – Donut Chart
A donut chart will be used to visualize performance distribution by target gender from the ads table.

- The displayed metric, such as **Impressions, Clicks, or Purchases**, will change dynamically based on the selected parameter.
- Each segment of the donut will represent a gender category.
- **Purpose:** To identify which gender segment contributes the most to the selected performance metric.

2. Target Age Group – Bar Chart
A bar chart will display performance across different target age groups defined in the ads table.

- Each bar will represent a specific age group.
- The metric shown will update dynamically based on the selected parameter.
- **Purpose:** To highlight which age group is most responsive to the campaigns and contributes the highest value for the chosen metric.

3. Country – Map Visualization
A map visualization will present performance by country using data from the users table.

- Bubble size or color intensity will represent the selected metric.
- The metric will change dynamically based on user selection.
- **Purpose:** To provide a geographic view of campaign reach, engagement, and performance across different countries.

4. Calendar Month – Calendar Heat Map
A calendar heat map will illustrate performance trends at the monthly level, based on the timestamp field in the ad_events table.

- Darker shades will indicate higher activity or stronger performance.
- Lighter shades will indicate lower activity.
- **Purpose:** To detect seasonal trends, identify peak-performing months, and spot periods of lower campaign activity.

5. Weekly Trend – Stacked Column Chart by Ad Type
A stacked column chart will show weekly performance trends segmented by ad type.

- **X-axis:** Week number from the **Date Table** linked to **ad_events**
- **Y-axis:** Selected metric
- **Stack segments:** Different **ad_type** values from the **ads** table
- **Purpose:** To compare how different ad types contribute to overall performance across weeks.

6. Hourly Trend – Area Chart

An area chart will visualize activity by hour of the day using ad_events[time_of_day].

- **X-axis:** Hour of the day (0–23)
- **Y-axis:** Selected metric
- The metric will update dynamically based on the chosen parameter.
- **Purpose:** To understand user engagement patterns throughout the day and identify peak activity hours.

7. Ad Type – Matrix

A matrix visualization will display the selected metric across ad types, with an additional breakdown by platform or other campaign dimensions.

- **Rows:** Ad Types
- **Columns:** Platforms (such as Facebook and Instagram) or other campaign-level dimensions
- **Values:** Selected metric
- **Purpose:** To compare performance across ad formats and platforms side by side in a structured view.